sophistry that there were four people with such blood types walking around Los Angeles) and adding to it the fact that he was the husband of the person and that there was additional evidence, then (owing to the compounding effect) the odds against him rise to several trillion trillion.

"Sophisticated" people make worse mistakes. I can surprise people by saying that the probability of the joint event is lower than either. Behavioral economists submitted rational and educated people (graduate students) to tests where they needed to produce the probability of a young woman with a liberal arts education being a *bank teller* or a *feminist bank teller*. They assigned on average a higher probability to her being a feminist bank teller than to that of her being a bank teller. I am glad to be a trader taking advantage of people's biases but I am scared of living in such a society.

AN ABSURD WORLD

Kafka's prophetic book, *The Trial*, about the plight of a man, Joseph K., who is arrested for a mysterious and unexplained reason hit a spot as it was written before we heard of the methods of the "scientific" totalitarian regimes. It projected a scary future of mankind wrapped in absurd self-feeding bureaucracies, with spontaneously emerging rules subjected to the internal logic of the bureaucracy. It spawned an entire literature of the *absurd*; the world may be too incongruous for us. I am terrified of certain lawyers. After listening to statements during the O.J. trial (and their effect) I was scared, truly scared of the possible outcome; my being arrested for some reason that made no sense probabilistically, and having to fight some glib lawyer in front of a randomness illiterate jury.

We said that mere judgment would probably suffice in a primitive society. It is easy for a society to live without mathematics – or traders to trade without quantitative methods – when the space of possible outcomes is one dimensional. One dimensional means that we are looking at one sole variable, not a collection of separate events. The price of one security is one dimensional, whereas the collection of the prices of several securities is multidimensional and requires mathematical modeling – we cannot easily see the collection of possible outcomes of the portfolio with a naked eye, and cannot even represent it on a graph as our physical

world has been limited to visual representation in three dimensions only. We will argue later why we run the risk of having bad models (admittedly, we have) or making the error of condoning ignorance – swinging between the Carybde of the lawyer who knows no math to the Scylla of the mathematician who misuses his math because he does not have the judgment to select the right model. In other words, we will have to swing between the mistake of listening to the glib nonsense of a lawyer who refuses science and that of applying the flawed theories of some economist who takes his science too seriously. The beauty of science is that it makes an allowance for both error types. Luckily, there is a middle road – but sadly, it is rarely traveled.

KAHNEMAN AND TVERSKY

Who are the most influential economists of the century, in terms of journal references, their followings, and their influence over the profession? No, it is not John Maynard Keynes, not Alfred Marshall, not Paul Samuelson, and certainly not Milton Friedman. They are Daniel Kahneman and Amos Tversky, psychology researchers whose specialty was to uncover areas where human beings are not endowed with rational thinking and optimal economic behavior.

The pair taught us a lot about the way we perceive and handle uncertainty. Their research, conducted on a population of students and professors in the early 1970s, showed that we do not correctly understand contingencies. Furthermore, they showed that in the rare cases when we understand probability, we do not seem to consider it in our behavior. Since the Kahneman and Tversky results, an entire discipline called *behavioral finance and economics* has flourished. It is in open contradiction with the orthodox so-called *neoclassical economics* taught in business schools under the normative names of *efficient markets*, *rational expectations*, and other such concepts. It is worth stopping, at this juncture, and discussing the distinction between normative and positive sciences. A *normative* science (clearly a self-contradictory concept) offers prescriptive teachings; it studies how things *should* be. Some economists, for example, (those of the efficient market religion) believe that humans are rational and act rationally